Department of Information Technology
Academic Year 2025 - 2026 (Even Semester)

Semester, Degree & Branch: VI Semester, B.Tech. IT
Course Code & Title: CCS354 & Network Security
Name of the Faculty member (s): Mrs. G.Sivasathiya, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit II / Kerberos
- **Course Outcome:** CO2
- **Topic Learning Outcome: TLO7**
- **Activity Chosen:** Peer Teaching

- **Justification:**
  Kerberos is a robust network authentication protocol designed to provide strong, mutual authentication between clients and servers over unsecured networks using secret-key cryptography. Peer teaching activity enhances the learning by boosting student engagement, confidence, and critical thinking through collaborative, personalized instruction. This type of teaching increases performance.

- **Time Allotted for the Activity:** 25 Minutes

- **Details of the Implementation:**
  - A topic Kerberos was chosen to be given as peer teaching topic. One student J.S Divya Sri shown willingness to learn and teach the topic to her peer-mates.
  - Divya Sri prepared a presentation in the topic Kerberos and its significances and she explained that in classroom environment.
  - This method created a collaborative environment where other students learn from one student through structured activities.
  - Fig 1.a shows the topic Kerberos is being explained by J.S. Divya Sri to her peers.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	3	2	2	2	2	1	2	1	2	3	1

(1 – Low 2 – Moderate 3 – High)

PO/PSO Mapped:

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO2	3	3	2	2	2	2	1	2	1	2	3	1
Justification for Correlation	Understanding cryptography concepts requires a solid foundation in engineering principles to grasp the technical aspects of security protocols, threat analysis, and protective measures.	Analyzing the need for encryption techniques involves identifying potential threats and vulnerabilities in network systems, which is essential for effective risk management and protection	Recognizing the need for cryptography involves considering security measures and algorithm during the design and development phases to ensure the delivery of safe and resilient solutions.	Cryptography often involves analyzing security algorithms and identifying potential weaknesses, which requires analytical skills and problem-solving abilities	Understanding the need for cryptography involves familiarity with modern encryption algorithms and tools to secure data and enhance privacy.	Effective cryptography requires collaboration within teams and individual contributions to design and implement encryption algorithms, ensuring robust data security and privacy	Communicating the significance of cryptography and its impact on data security to stakeholders is essential for ensuring alignment and garnering support for encryption initiatives	Understanding the need for cryptography involves considering its implications on project timelines and budgets, necessitating effective project management and financial planning for secure system development	Recognizing the dynamic nature of cryptography, engineers should cultivate a mindset of continuous learning to stay updated with evolving encryption techniques and security technologies	Apply the knowledge of proficiency in cryptography and provide the foundation to ensuring robust encryption of complex computational problems.	Apply the knowledge of cryptography to provide solutions to emerging technologies such as mobile application security, ensuring the protection of sensitive data and providing reliable IT solutions	Developing cryptographic solutions is a base for implementing a project as an ethical software engineer, ensuring the secure handling of data and protecting user privacy.

- **Images / Screenshot of the practice:**

Figure 1. b Kerberos topic explained by J.S.Divya Sri

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***
  - Students felt more confident when sharing ideas and asking questions in front of their peers.
  - Educators observed increased engagement, collaboration, and responsibility among students during the activity.
- ***Benefit of the practice:***
  - Students were actively participated in this activity rather than remaining passive listeners.
  - Teaching peers reinforced the student teachers' own understanding and improved knowledge retention.
  - The activity promoted collaborative learning and mutual support among students.
- ***Challenges faced in implementation:***
  - Time management was challenging, especially when discussions extended beyond the planned schedule.
  - Some students preferred traditional teacher-led instruction and initially resisted peer-led learning
- ***References:***

  - https://www.wgu.edu/blog/peer-learning2208.html
  - https://cognitivecardiomath.com/cognitive-cardio-blog/peer-teaching-overview-benefits-and-strategies/

Signature of Faculty Member **HOD**